

PROJECT PROGRESS FOR RESEARCH PROJECT

CST 498-6 / IIT 474-6

Bilingual Voice Transcription and Synthesis Preserving Speaker Identity Using NLP and Deep Learning

R.Nilakshan

UWU/CST/21/055

Computer Science and Informatics

I.R.Janukshan

UWU/CST/21/056

Computer Science and Informatics

P.Piranya

UWU/CST/21/065

Computer Science and Informatics

Department of Computer Science and Informatics

Faculty of Applied Sciences

Uva Wellassa University

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Declaration

We, hereby, declare that the research proposal titled “ Bilingual Voice Transcription and Synthesis Preserving Speaker Identity Using NLP and Deep Learning” is our original work and has been conducted solely for the purpose of fulfilling the partial fulfilment of the requirement of the course unit CST 498-6, Research Project for the degree of Computer Science and Technology.

We affirm that this work is free from any form of plagiarism and has not been submitted, in part or in full, for any other course, program, or qualification. All sources used in this proposal have been properly cited and acknowledged.

Student Details:

Enrolment No.	Name with Initials	Email Address	Signature
UWU/CST/21/055	R.Nilakshan	cst21055@std.uwu.ac.lk	
UWU/CST/21/056	I.R.Janukshan	cst21056@std.uwu.ac.lk	
UWU/CST/21/065	P.Piranya	cst21065@std.uwu.ac.lk	

Primary Supervisor Details:

Name with Initials	Email Address	Signature
E.M.U.W.J.B.Ekanayake	jayalath@uwu.ac.lk	

Co-Supervisor Details:

Name with Initials	Email Address	Signature
H.P.D.P.Pathirana	dimuth@uwu.ac.lk	

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List of Abbreviations

Artificial Intelligence	AI
Text-to-Speech	TTS
Automatic speech recognition	ASR
Neural Machine Translation	NMT
Signal-to-Noise Ratio	SNR
Perceptual Evaluation of Speech Quality	PESQ
Word Error Rate	WER
Short-Time Objective Intelligibility	STOI
Bilingual Evaluation Understudy	BLEU

1 Introduction

1.1 Project Background

Bilingual and multilingual speech synthesis has emerged as one of the most dynamic research areas in modern artificial intelligence because of its vast applications in human-computer interaction, automated dubbing, multilingual content creation, and assistive communication technologies. Traditional TTS systems were mainly developed to generate intelligible speech; however, they often lacked naturalness, emotional expressiveness, and speaker-identity preservation across languages. With globalization driving the demand for cross-lingual communication, the ability to synthesize natural, emotionally expressive, and speaker-consistent bilingual speech has become increasingly important.

Recent advancements in NLP and Deep Learning have significantly enhanced the capabilities of speech synthesis systems. Early statistical and rule-based models struggled with prosody modeling, emotion transfer, and linguistic variability, often resulting in mechanical or monotone speech. Modern neural architectures, however, have enabled end-to-end modeling of acoustic and linguistic representations directly from data. Nuthakki et al. (2023) demonstrated that multi-feature fusion methods can effectively improve multilingual speech synthesis accuracy and quality, even in low-resource language settings. Similarly, Ambadekar et al. (2025) conducted a comprehensive review of multilingual voice-to-voice translation frameworks and emphasized the need to integrate translation, synthesis, and speaker-preserving modules to achieve natural and emotionally engaging cross-language communication.

In addition to linguistic accuracy, researchers have recognized the critical importance of prosody and paralinguistic features in producing expressive and human-like speech. Kiss-Vetráb & Gosztolya (n.d.) emphasized that computational paralinguistics-including intonation, rhythm, and emotion is essential for generating speech that conveys affective meaning and natural human expressiveness. Building on this perspective, Klein & McCartney (2025) introduced a cross-lingual prosody-preservation framework capable of maintaining the pitch, rhythm, and stress patterns of the source language in the target-language output, thus ensuring higher perceptual realism and continuity of speaker style.

Furthermore, emotion-aware voice conversion has become a pivotal extension of multilingual synthesis. Yang (2025) explored deep emotional text-to-speech and voice-conversion models that can transfer affective tone and personality between voices while maintaining lin-

guistic content. These approaches have shown significant potential for applications such as film dubbing, interactive media, and personalized virtual assistants, where emotional fidelity is a key factor for audience engagement.

Together, these recent studies demonstrate that current research in multilingual speech synthesis is advancing toward deep, emotion-aware, and prosody-preserving systems. The integration of deep learning, NLP, and paralinguistic modeling forms the foundation for developing bilingual voice transcription and synthesis frameworks capable of preserving speaker identity and emotional depth across languages, which is the central focus of the present study.

1.2 Problem Definition

Although substantial progress has been achieved in deep learning-based multilingual speech synthesis, existing systems still face significant challenges in maintaining a balance between **linguistic accuracy, naturalness, and speaker identity preservation** when generating bilingual or multilingual speech. Many state-of-the-art Text-to-Speech (TTS) models are capable of producing intelligible speech in multiple languages; however, they often fail to retain the **emotional tone, speaking style, and prosodic characteristics** of the original speaker, especially when translating between linguistically and phonetically distant languages.

Traditional TTS frameworks primarily focused on improving intelligibility and fluency, often at the expense of expressiveness and emotion. While recent studies have integrated multi-feature fusion and deep neural architectures to enhance synthesis quality, persistent gaps remain in emotional and prosodic transferability. Nuthakki et al. (2023) demonstrated that multilingual models using multi-feature fusion techniques improve cross-lingual synthesis but still encounter challenges in preserving unique speaker traits and affective dynamics. Likewise, Ambadekar et al. (2025) noted that many multilingual translation systems lack sufficient alignment between linguistic and emotional layers, resulting in monotonous and context-insensitive speech outputs.

Another limitation lies in the treatment of prosody and paralinguistic cues. Studies such as Kiss-Vetráb & Gosztolya (n.d.) emphasize that prosodic and paralinguistic elements-such as rhythm, pitch, and emotions-are vital to maintaining authenticity in synthesized speech, yet they are frequently underrepresented in current bilingual synthesis models. Similarly, Klein & McCartney (2025) highlighted that even advanced prosody-preservation frameworks still strug-

gle to maintain consistent rhythmic and tonal mapping across languages, reducing perceived naturalness and continuity of speech identity. Furthermore, Yang (2025) identified that deep emotional text-to-speech and voice-conversion architectures, while effective at transferring affective tone within a single language, still face difficulty when scaling emotion preservation across multiple languages and voices.

In summary, the central problem is that existing bilingual and multilingual speech synthesis systems are not yet capable of fully preserving both **speaker individuality** and **emotional expressiveness** across languages. This limitation diminishes the authenticity and naturalness of synthesized voices, particularly in applications such as film dubbing, virtual assistants, and accessibility technologies where speaker identity and emotional realism are essential. Therefore, there is a critical research need to develop an advanced bilingual voice transcription and synthesis model that integrates deep learning, NLP-based translation, and prosody-preserving emotional modeling to generate expressive, identity-consistent multilingual speech.

1.3 Project aims/objectives

Aim:

The aim of this study is to develop a **bilingual voice transcription and synthesis system (Spanish and English)** that accurately translates speech across languages while preserving the speaker's vocal identity, prosody, and emotional tone. By leveraging recent advancements in **deep learning, multilingual Natural Language Processing (NLP)**, and **prosody-preserving synthesis**, the system seeks to enhance the naturalness, expressiveness, and authenticity of multilingual voice output. This research aims to contribute to more inclusive, emotionally adaptive, and human-like speech-based applications such as film dubbing, virtual assistants, and accessibility technologies Nuthakki et al. (2023), Ambadekar et al. (2025), Klein & McCartney (2025), Kiss-Vetráb & Gosztolya (n.d.), Yang (2025).

Objectives:

- To design and implement a voice-to-voice translation framework for Spanish to English that preserves the speaker's voice characteristics and emotional tone using pre-trained ASR, NMT, and TTS models.(Bidirectional) Nuthakki et al. (2023).
- To ensure temporal alignment between original and translated speech by applying speaker diarization, pause-based segmentation, and timing control for natural synchronization.
- To integrate the synthesized dialogue with the original background audio and video, creating an automated dubbing system that produces complete translated video outputs with minimal human intervention.

2 Project Description

The proposed research aims to develop a bilingual (**Spanish and English**) voice transcription and synthesis system capable of accurately translating spoken content between two languages while preserving the speaker’s vocal identity, prosody, and emotional tone. This initiative is grounded in the need for natural and expressive multilingual speech systems, particularly in domains such as film dubbing, assistive communication, and international media production, where authenticity and emotional fidelity are critical. Unlike conventional TTS systems that often lose speaker-specific characteristics and expressive qualities in cross-language transitions, the proposed framework seeks to maintain emotional depth and speaker individuality by leveraging advancements in deep learning, NLP, and voice synthesis technologies.

Recent literature supports the growing importance of emotional and prosodic consistency in multilingual speech applications. According to Nuthakki et al. (2023) Nuthakki et al. (2023), deep neural networks enhanced with multi-feature fusion methods have significantly advanced multilingual TTS accuracy.

However, challenges persist in preserving emotional and prosodic nuances when transferring speech across languages. Ambadekar et al. (2025) Ambadekar et al. (2025) further emphasize the necessity of integrating translation and synthesis modules into a single cohesive system for natural voice-to-voice translation, pointing out that modular or disjointed systems often result in robotic or monotonous output.

Similarly, Klein and McCartney (2025) Klein & McCartney (2025) stress the significance of cross-lingual prosody preservation, especially in applications where the rhythm, tone, and stress of speech contribute significantly to the speaker’s expressiveness.

To address these gaps, the proposed system will consist of five tightly integrated components. First, the speech transcription module will employ state-of-the-art deep learning models (such as Transformer-based ASR) to convert bilingual audio input into accurate textual transcriptions. These models will be trained and fine-tuned on large multilingual datasets, ensuring robust recognition across accents, speaking rates, and noisy conditions.

Second, a cross-lingual NLP translation engine will be used to convert the transcribed source-language text into the target language. This engine will not only translate literal content but also attempt to preserve idiomatic expressions, cultural context, and emotional subtext. Pre-trained models like MarianMT will be fine-tuned for the domain-specific dialogue of films and conver-

sational speech to improve contextual translation fidelity.

The third stage involves prosody and paralinguistic modeling. Building on the findings of Kiss-Vetráb and Gosztolya Kiss-Vetráb & Gosztolya (n.d.), this module will analyze speech features such as pitch contour, duration, energy levels, and intonation patterns to capture the speaker's expressive behavior. These features will be encoded and transferred into the synthesis process to preserve emotional tone and rhythm during speech generation in the target language.

The fourth module centers on emotional and identity-aware voice synthesis. Using architectures like YourTTS or Tacotron-Glow-TTS in conjunction with HiFi-GAN vocoders, the system will clone the original speaker's voice and infuse it with emotional and prosodic cues identified earlier. Inspired by the approach of Yang (2025) Yang (2025), the model will ensure that the synthesized voice not only matches the speaker's timbre but also conveys the same intensity of emotion, making the speech output sound natural and empathetic.

The final module is responsible for synchronization and evaluation. Once the synthesized speech is generated, it will be aligned with the video footage or original audio context using forced alignment algorithms to maintain lip-sync accuracy. The entire pipeline will be evaluated using objective metrics such as WER, PESQ, and STOI, alongside subjective feedback from human evaluators through listener surveys. Participants will rate samples on emotional expressiveness, voice similarity, naturalness, and overall quality using a Likert-scale approach. Together, these components contribute to a unified, end-to-end bilingual speech system that is both technically robust and perceptually convincing. By integrating deep neural ASR, contextual translation, prosody modeling, emotional synthesis, and rigorous evaluation, the system aims to bridge the gap between machine-generated speech and human expressiveness. This research stands to significantly advance the field of multilingual TTS by offering a novel framework that respects the speaker's vocal signature and emotional expressiveness across languages, potentially transforming applications in education, entertainment,

3 Literature Review

Recent advances in multilingual voice-to-voice translation have demonstrated significant improvements in both speech recognition and synthesis. Ambadekar et al. Ambadekar et al. (2025) provide a comprehensive review of multilingual voice-to-voice translation systems, emphasizing accurate language alignment and prosody preservation for intelligibility across languages. Kiss-Vetráb and Gosztolya Kiss-Vetráb & Gosztolya (n.d.) highlight the role of computational paralinguistics, stressing that effective audio analysis and feature extraction methodologies are crucial for reliable speech processing, particularly in multilingual contexts. Klein and McCartney Klein & McCartney (2025) investigate cross-lingual prosody preservation in speech generation applications, showing that maintaining natural intonation and rhythm enhances user comprehension and engagement. Nuthakki et al. Nuthakki et al. (2023) propose a deep learning approach for multilingual speech synthesis using multi-feature fusion methods, improving the quality and naturalness of synthesized voices in low-resource languages. Furthermore, Yang Yang (2025) explores deep emotional text-to-speech and voice conversion, providing insights into generating expressive speech while retaining speaker identity. Collectively, these studies underscore the importance of combining robust acoustic feature extraction, prosody modeling, and advanced deep learning techniques to achieve high-quality multilingual voice-to-voice translation and synthesis systems.

4 Methodology

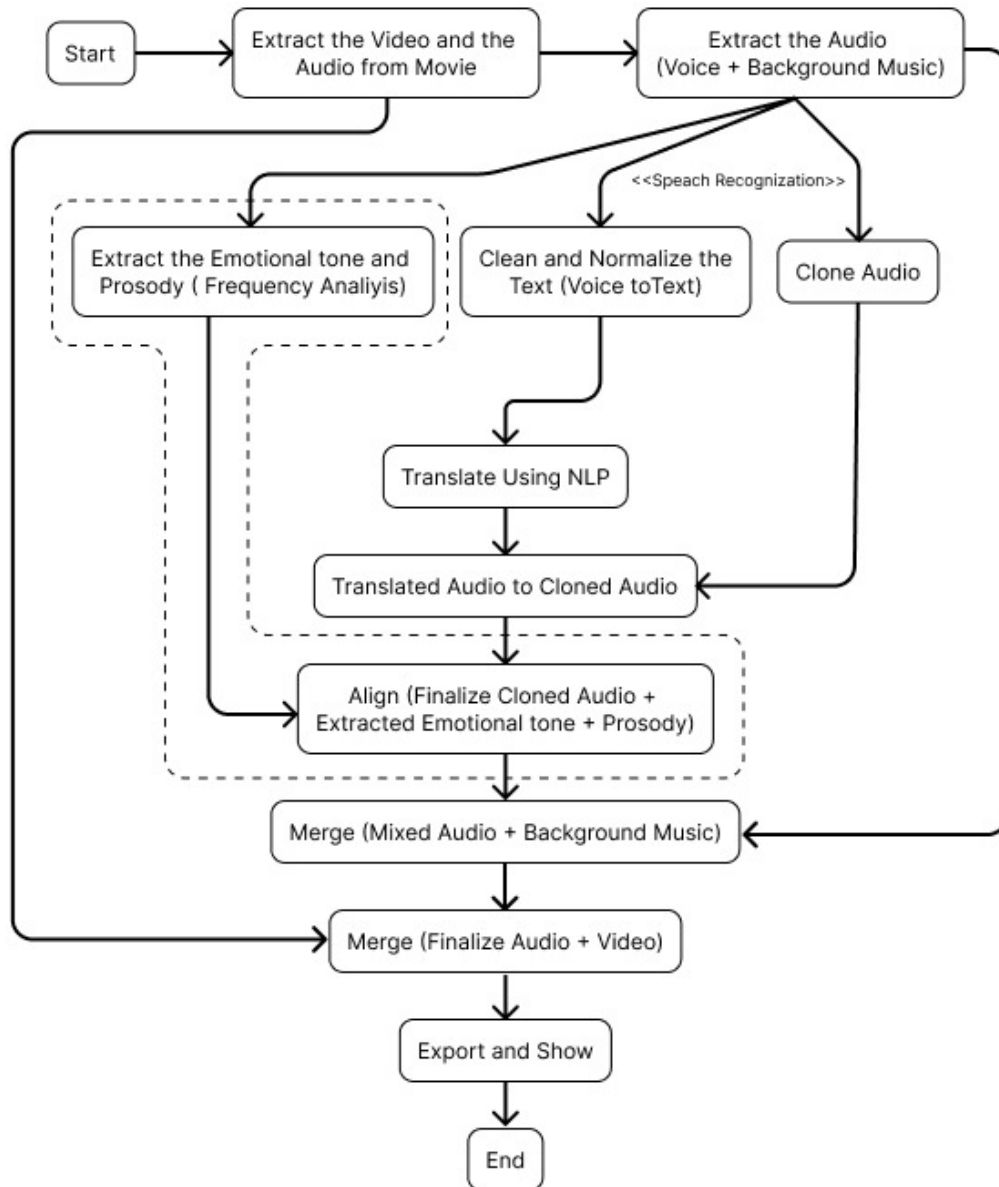


Figure 1: Methodology Diagram

4.1 Research Approach and Strategy

This study adopts a quantitative research approach supported by experimental design. The research is grounded in positivism, emphasizing measurable performance outcomes and objective evaluation through computational experimentation. The strategy involves developing a prototype multilingual voice translation system integrating ASR, NMT, and TTS modules conditioned on cloned voice features. Each component will be trained and tested using publicly available datasets, and results will be analyzed using standard quantitative metrics such as WER, SNR, and PESQ. In addition, a limited qualitative component will be included through human perceptual evaluation to assess the emotional authenticity and naturalness of the synthesized output, complementing the quantitative findings.

4.2 Data Collection Methods

The study employs both archival research and experimental data generation methods to ensure comprehensive data coverage and high-quality model training.

Archival Datasets (Secondary Data): Publicly available pre-recorded bilingual and multilingual speech datasets—considered as *secondary data*—such as *Mozilla Common Voice*, *VoxCeleb 1 & 2*, the *University of Edinburgh Speech Data Repository*, and the *CVSS – Common Voice Speech-to-Speech Corpus* will be utilized. These sources provide transcribed voice samples across multiple languages suitable for training models in ASR, NMT, and TTS synthesis. The data include speech recordings, textual transcriptions, and metadata describing speakers, languages, and acoustic characteristics.

In particular, the CVSS corpus (developed by Google in 2022) builds upon Mozilla Common Voice and CoVoST-2 to offer sentence-aligned Spanish→English speech pairs for approximately 87,000 utterances. The English translations are synthesized using TTS in two formats: **CVSS-C**, which uses a single neutral voice for all English outputs, and **CVSS-T**, which performs voice transfer to retain the original Spanish speaker’s vocal identity in the English synthesis. This enables clean parallel alignment and speaker-aware translation suitable for bilingual speech-to-speech systems.

Experimental Data Processing: Movie dialogue segments will be extracted, voice tracks isolated, and audio cleaning, normalization, and segmentation will be performed. These processed samples will be used to generate new training and validation subsets for emotional tone analysis, speaker embedding, and translation alignment tasks.

Supplementary Survey: A small-scale human evaluation survey will be conducted in which participants listen to translated and synthesized audio clips and rate them based on naturalness, emotional expressiveness, and speaker similarity using a five-point Likert scale. This combination of archival (secondary) and experimental (primary) methods ensures both technical robustness and perceptual validation of the developed system.

4.3 Sampling Strategy

(a) Dataset Sampling: A **purposive sampling** approach will be used to select bilingual speech data that meet the following criteria:

- High-quality recordings with clear articulation and minimal background noise.
- Balanced gender distribution and diversity in accents and speaking styles.
- Availability of accurate transcriptions in both source and target languages.

Approximately 15–20 minutes of speech per language will be selected for training and evaluation. The data will be divided into training (70%), validation (15%), and testing (15%) subsets to maintain performance consistency and prevent overfitting.

(b) Human Listener Sampling: Around 5-6 participants proficient in at least one of the selected languages will be recruited through convenience sampling. They will voluntarily participate in perceptual listening sessions and provide feedback on the naturalness and emotional quality of synthesized speech.

4.4 Data Analysis Techniques

This study adopts both **quantitative** and **qualitative** analysis methods to comprehensively evaluate system performance.

Quantitative Analysis: Objective evaluation metrics such as WER, PESQ, STOI, and speaker similarity scores will be computed.

Qualitative Analysis: Thematic analysis will be performed on participant feedback to identify recurring themes related to perceived emotional accuracy, voice consistency, and speech naturalness. Quantitative results and qualitative insights will then be cross-validated to derive a holistic understanding of system performance and listener perception.

4.5 Ethical Considerations

The research strictly adheres to institutional, national, and international ethical standards.

- **Informed Consent:** All participants will receive a detailed explanation of the study’s purpose, procedures, and voluntary nature prior to participation.
- **Data Privacy:** No personal or sensitive data will be collected. All datasets used are open-source or appropriately licensed.
- **Anonymity and Confidentiality:** Responses will be anonymized, and all records will be stored securely with restricted access.
- **Fair Use Compliance:** Movie clips and speech samples used for testing will comply with Creative Commons or educational fair-use terms.
- **Minimization of Risk:** Listening sessions will be short, non-intrusive, and monitored to avoid participant discomfort or fatigue.

4.6 Evaluation Strategy

The evaluation of the proposed system combines both **objective computational assessments** and **subjective perceptual testing** to validate performance comprehensively.

Objective Evaluation: Model outputs will be compared using established quantitative metrics such as WER, PESQ, STOI, and speaker-embedding similarity scores. Emotional tone preservation will be analyzed through acoustic feature correlation between original and synthesized speech samples.

Subjective Evaluation: Human evaluators will rate audio samples based on naturalness, emotional authenticity, and speaker similarity. The mean opinion scores and inter-rater reliability will be calculated to assess listener agreement.

Benchmark Comparison: Performance will be benchmarked against state-of-the-art bilingual TTS models to determine improvements in naturalness, emotional accuracy, and identity preservation.

Validation and Reproducibility: Cross-validation and repeated experiments on unseen test sets will be conducted to ensure result consistency, model generalizability, and reproducibility.

5 Results and Discussion

The primary focus was on preparing and standardizing the audio dataset to ensure consistent quality across all samples. English audio data were collected and organized into a structured directory format. All audio files were resampled to 16 kHz to match the input requirements of the Whisper ASR model. Noise reduction and amplitude normalization techniques were applied to minimize background interference and volume inconsistencies. These preprocessing steps were critical to ensure that ASR performance would not degrade due to variations in audio quality. Successful completion of this stage established a stable foundation for downstream processing.

Focused on segmenting continuous speech recordings into smaller, meaningful utterances using silence detection techniques. Long audio segments can negatively impact ASR accuracy due to context drift and accumulated transcription errors. By applying silence thresholds based on duration and energy levels, natural pauses in speech were identified. This process resulted in 103 segmented utterances, most of which ranged between one and three seconds in length. This segment duration is optimal for modern ASR systems and contributes to improved transcription reliability.

The segmented English audio data were transcribed using the Whisper ASR model. Each segment was processed independently to minimize error propagation. Transcription accuracy was evaluated using Word Error Rate (WER). The system achieved an average WER of approximately 0.079, indicating high transcription accuracy. Most errors were minor, such as missing articles or slight word substitutions, and did not significantly affect sentence meaning. This high-quality ASR output provided a reliable textual input for the translation stage.

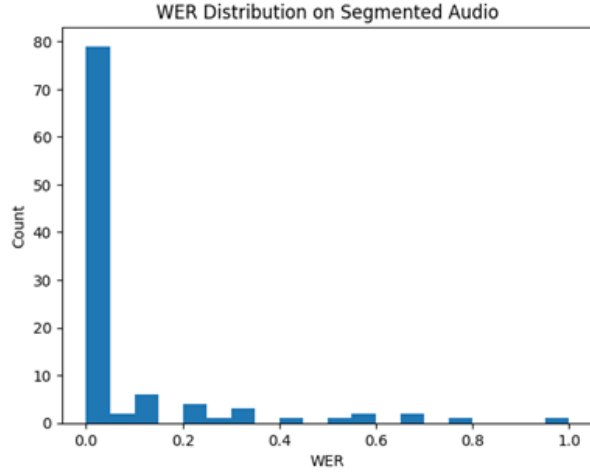


Figure 2: WER Distribution on Segmented Audio

To evaluate the quality of the English–Spanish translation produced by our system in the absence of direct Spanish reference transcripts, we employed a back-translation evaluation strategy. In this approach, the Spanish sentences generated by the model were translated back into English and compared with the original English text produced by the ASR component. We measured the closeness of the model outputs using the BLEU score, which quantifies the overlap between the back-translated text and the original text based on n-gram matching. Our model achieved a BLEU score of approximately 61, indicating a strong degree of semantic overlap and suggesting that the core meaning of the original sentences was preserved through the translation process. In addition to BLEU, we calculated a similarity metric defined as $(1 - \text{WER})$, where WER denotes word error rate, which yielded an average value of approximately 0.77. This further supports the observation that the translated texts closely resemble the original English content in terms of word sequence and structure. Together, these metrics demonstrate that the translation component of our system maintains effective meaning preservation even without direct Spanish ground truth references.

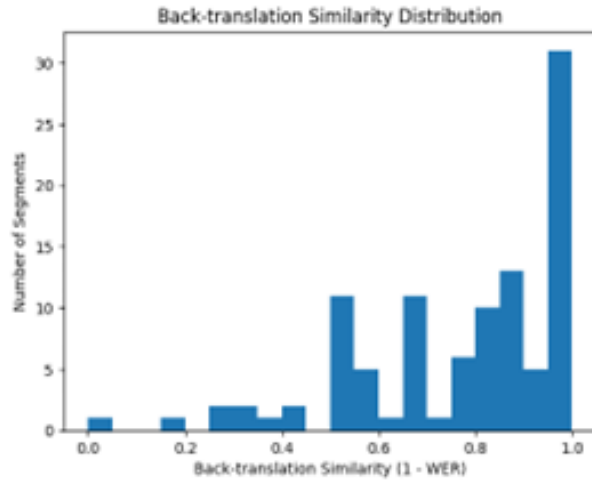


Figure 3: Back-translation Similarity Distribution

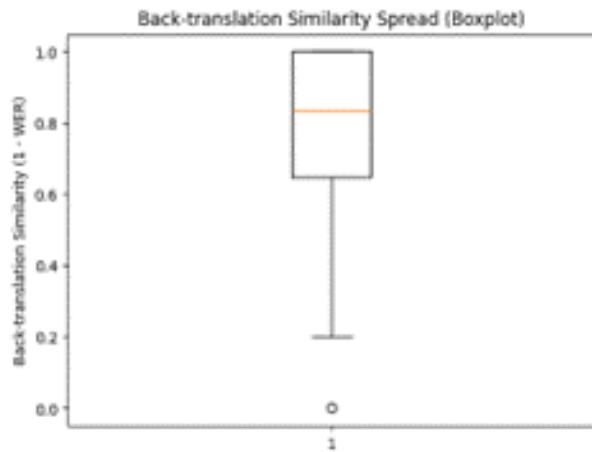


Figure 4: Back-translation Similarity Speed

Comparative Analysis of Training Approaches Two training approaches were analyzed. The first approach utilized the Opus Books dataset containing paired English and Spanish audio. This method attempted to directly adapt the trained model to the project's English audio dataset. However, performance was limited due to domain mismatch and lack of alignment with real-world speech characteristics. The second approach followed a cascaded pipeline: English audio $\rightarrow$ English text $\rightarrow$ Spanish text. This approach provided better control, clearer evaluation metrics, and higher robustness. Quantitative evaluation results clearly favored this method.

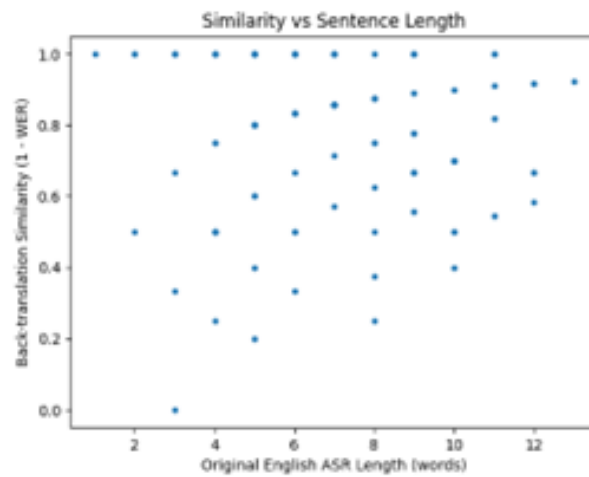


Figure 5: Original English ASR Leng



Figure 6: NMT Training Loss

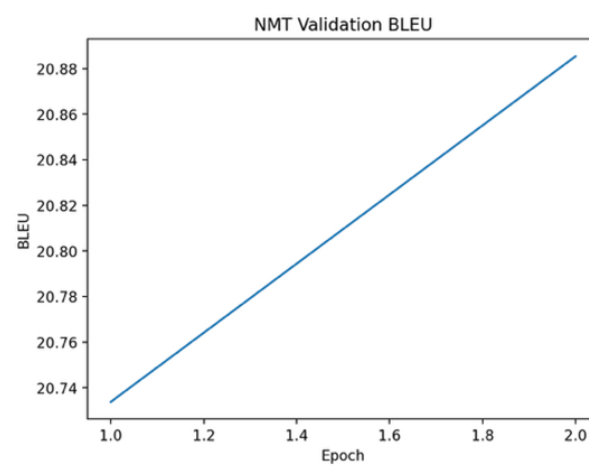


Figure 7: NMT Validation Using BLEU

Day 6 - Final Model Selection and Discussion Based on quantitative results and qualitative observations, the cascaded ASR-to-NMT pipeline was selected as the final model. The combination of low ASR WER, high BLEU score, and strong similarity metrics demonstrated reliable semantic preservation. The final system successfully balances performance, interpretability, and scalability. The results confirm that BLEU-based back-translation is an effective evaluation strategy when direct reference translations are unavailable, making the proposed approach suitable for real-world bilingual speech translation applications.

6 Resources

6.1 Hardware resources

- **High-Performance GPU Workstation:** NVIDIA RTX 3060/4060 Ti or higher with at least 8–12 GB VRAM for deep-learning training and inference.
- **CPU:** Intel Core i7 / AMD Ryzen 7 or equivalent, minimum 8 cores for preprocessing and general tasks.
- **Memory (RAM):** Minimum 16 GB, recommended 32 GB for smooth data handling.
- **Storage:** SSD with at least 512 GB capacity to store bilingual audio datasets, model checkpoints, and output files.
- **Audio Devices:** Studio-quality headphones or speakers for subjective evaluation of synthesized speech.

6.2 Software resources

- **Operating System:** Ubuntu 22.04 LTS or Windows 11 Pro for compatibility with deep-learning frameworks.
- **Programming Languages:** Python 3.10+ and JavaScript (React / Next.js frontend).
- **Development Environment:** Visual Studio Code, Jupyter Notebook, and Google Colab Pro for cloud-based training.
- **Database Management:** MongoDB Atlas or MySQL for storing processed data and metadata.
- **Version Control:** Git and GitHub for collaborative development and code tracking.
- **Project Documentation:** Overleaf / LaTeX for formal report preparation.

6.3 Libraries and Frameworks

- **Deep Learning:** TensorFlow 2.x, PyTorch 2.x, and Keras API for model development, training, and inference.

- **Speech Recognition & NLP:** OpenAI Whisper, DeepSpeech, Transformers (Hugging Face), spaCy, and NLTK for transcription, translation, and language modeling tasks.
- **Voice Cloning & TTS:** YourTTS, Tacotron 2, Glow-TTS, HiFi-GAN, and Coqui TTS for multilingual speech synthesis while preserving speaker identity.
- **Audio Processing:** Librosa, PyDub, SoundFile, SciPy, and FFmpeg-Python for signal processing, feature extraction, and audio manipulation.
- **Evaluation Metrics:** PESQ, STOI, WER, SNR, and Cosine Similarity for objective performance analysis and speaker embedding validation.
- **Data Visualization & Analysis:** Matplotlib, Seaborn, Pandas, NumPy, and Scikit-Learn for data visualization, preprocessing, and statistical evaluation.

6.4 Tools and Platforms

- **Google Colab Pro / Kaggle Kernels:** For GPU-accelerated model training and experimentation in cloud environments.
- **Anaconda:** For environment management, dependency handling, and package isolation in Python projects.
- **GitHub Projects:** For issue tracking, version control, and collaborative project documentation.
- **Render / Vercel / Hugging Face Spaces:** For model deployment, API hosting, and demonstration of synthesized voice systems.
- **Figma / Draw.io:** For creating the methodology workflow diagrams, posters, and UI design illustrations.
- **Audacity:** For audio editing, trimming, and background noise reduction in collected datasets.
- **MS Excel / Google Sheets:** For organizing, analyzing, and visualizing quantitative evaluation and survey results.

7 Individual Contribution and Work Plan

UWU/CST/21/055	UWU/CST/21/056	UWU/CST/21/065
Pull video/audio, organize datasets, metadata sheets.		
Source separation, denoise, normalization, segmenting dialogues.		
Train/finetune speaker encoders; clone target actor voice.	Choose ASR model, punctuation/diacritics restore, text cleanup.	
Feature extraction pipelines; prosody vectors for conditioning.		
Conditioned TTS inference/training; latency		Conditioned TTS inference/training; latency
Source separation, denoise, normalization, segmenting dialogues.		Source separation, denoise, normalization, segmenting dialogues.
	Loudness match, ducking, scene-wise mixdown	
Replace audio track, render deliverables, versioning.		
Metric scripts, baselines, ablations.		
	Survey design, IRB/consent text, analysis and visualization.	
Write-ups, figures, repo README, presentation slides/poster.		Write-ups, figures, repo README, presentation slides/poster

Table 1: Individual Contribution Table

8 Project Timeline

The project is planned to be completed through a structured series of objectives and development phases over approximately **40 weeks**. Each objective corresponds to a major research milestone in the bilingual voice transcription and synthesis framework. The phases include data preparation, model development, evaluation, and documentation, ensuring smooth and progressive implementation.

- **Objective 1: Data Collection and Preprocessing** This phase focuses on collecting bilingual speech data, extracting audio from movie clips, and preparing the dataset for model training.

- **Phase 1: Dataset Preparation and Cleaning**

- * Weeks 1–3: Identify and download open-source datasets such as Mozilla Common Voice, VoxCeleb, and the University of Edinburgh Speech Repository.
- * Weeks 4–6: Extract movie dialogue segments, separate voice from background, normalize audio, and annotate data for transcription and translation.

- **Objective 2: Speech Transcription and Translation Module** Develop an automatic speech-to-text (ASR) model and integrate neural translation for bilingual language conversion.

- **Phase 2: ASR and NLP Translation Development**

- * Weeks 7–10: Train and fine-tune the ASR model using deep learning architectures for bilingual transcription.
- * Weeks 11–13: Implement neural machine translation (NMT) to convert the transcribed text between source and target languages.

- **Objective 3: Prosody Analysis, Emotion Modeling, and Voice Cloning** This phase handles the modeling of pitch, tone, rhythm, and emotion, followed by cloning the speaker’s unique voice.

- **Phase 3: Emotional and Prosody Feature Extraction**

- * Weeks 14–17: Analyze emotional and prosodic features (pitch, rhythm, and tone) from the extracted voice samples.

- **Phase 4: Speaker Embedding and Voice Cloning**
 - * Weeks 18–22: Develop and train voice cloning models using speaker-embedding techniques to preserve identity.
- **Objective 4: Speech Synthesis and Audio Integration** Implement the text-to-speech (TTS) synthesis process using the cloned voice and align the generated audio with the original video content.
 - **Phase 5: Emotion-Aware Speech Synthesis**
 - * Weeks 23–28: Generate translated audio using emotion-aware TTS models (Tacotron 2, YourTTS, HiFi-GAN).
 - **Phase 6: Audio Alignment and Video Integration**
 - * Weeks 29–32: Align synthesized audio with original video frames to ensure lip synchronization and timing accuracy.
- **Objective 5: Evaluation, Testing, and Documentation** The final stage focuses on evaluating model performance, validating emotional and speaker consistency, and completing documentation.
 - **Phase 7: Evaluation and Validation**
 - * Weeks 33–37: Conduct objective testing (WER, PESQ, STOI, similarity) and subjective human evaluation surveys.
 - **Phase 8: Documentation and Final Report**
 - * Weeks 38–40: Compile research documentation, finalize report writing, and prepare presentation materials.

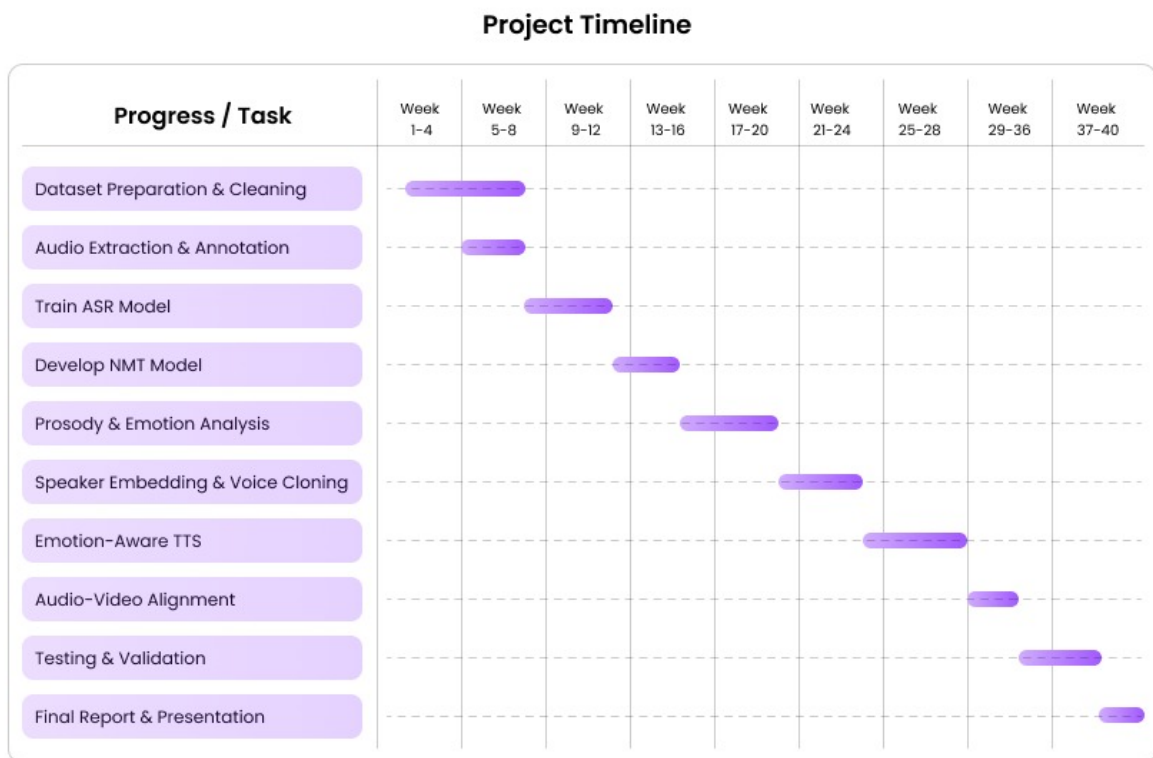


Figure 8: Planned project timeline for bilingual voice transcription and synthesis system.

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